OPTICAL NETWORKS

Lecture 10

Numerical-1

Wavelength Standards

Consider a passive WDM link of length L, consisting of single-mode fiber into which five wavelengths are launched through an optical combiner such that the aggregate launch power at its output is 5 mW. These five wavelengths are *centered* on the 193.1 THz ITU grid, with uniform 100 GHz interchannel spacing. The transmitters all use directly modulated DFB lasers with a spectral width of 0.1 nm. At the end of this link, the channels are optically demultiplexed and are each received by a direct detection *pin* receiver. For this problem, neglect all losses and crosstalk associated with the WDM mux and demux.

(a) Calculate the wavelengths (in nm, to two decimal places) associated with these five channels.

(b) Draw and label a diagram illustrating this configuration.

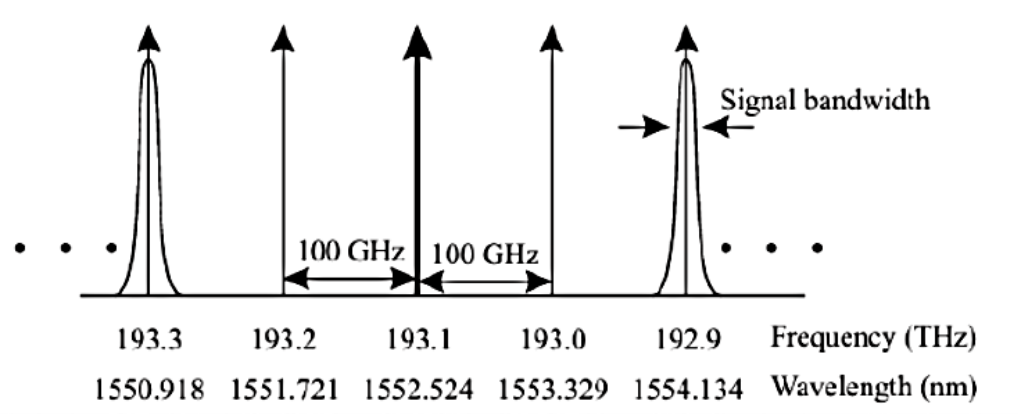
Recall:

- *The wavelengths and frequencies used in WDM systems have been standardized on a frequency grid by the International Telecommunications Union (ITU).*
- *It is an infinite grid centered at 193.1 THz*

Numerical-1

Given center frequency 193.1 THz
Channel spacing: 100GHz =0.1THz
* Velocity of light =2.9979 x 10^8 m/s

Freq (THz)	Wavelength (nm)
192.9	1554.1340487
193.0	1553.3287979
193.1	1552.5243811
193.2	1551.7207971
193.3	1550.9180445



Stimulated Brillouin scattering

- Stimulated Brillouin scattering (SBS) may be regarded as the modulation of light through thermal molecular vibrations within the fiber. The scattered light appears as upper and lower sidebands which are separated from the incident light by the modulation frequency. The incident photon in this scattering process produces a phonon of acoustic frequency as well as a scattered photon. This produces an optical frequency shift which varies with the scattering angle because the frequency of the sound wave varies with acoustic wavelength.
- The frequency shift is a maximum in the backward direction, reducing to zero in the forward direction, making SBS a mainly backward process.
- Brillouin scattering is only significant above a threshold power density. Assuming that the polarization state of the transmitted light is not maintained, the threshold power P_B is given by:

$$P_B = 4.4 \times 10^{-3} d^2 \lambda^2 \alpha_{dB} \nu \text{ watts}$$

where d and λ are the fiber core diameter and the operating wavelength, respectively, both measured in micrometers, α_{dB} is the fiber attenuation in decibels per kilometer and ν is the source bandwidth (i.e. injection laser) in gigahertz. The expression given in above equation allows the determination of the threshold optical power which must be launched into a single-mode optical fiber before SBS occurs.

Ref: "Optical Fiber Communications Principles and Practice"- John M. Senior, 3rd Edition.

Stimulated Raman scattering

- Stimulated Raman scattering (SRS) is similar to SBS except that a high-frequency optical phonon rather than an acoustic phonon is generated in the scattering process.
- sAlso, SRS can occur in both the forward and backward directions in an optical fiber, and may have an optical power threshold of up to three orders of magnitude higher than the Brillouin threshold in a particular fiber.
- Using the same criteria as those specified for the Brillouin scattering threshold, the threshold optical power for SRS P_R in a long single-mode fiber is given by:

$$P_R = 5.9 \times 10^{-2} d^2 \lambda \alpha_{dB} \text{ watts}$$

where d, λ and α_{dB} are as specified for SBS threshold.

Ref: "Optical Fiber Communications Principles and Practice"- John M. Senior, 3rd Edition.

Numerical-2

A long single-mode optical fiber has an attenuation of 0.5 dB km^{-1} when operating at a wavelength of 1.3 μm. The fiber core diameter is 6 μm and the laser source bandwidth is 600 MHz. Compare the threshold optical powers for stimulated Brillouin and Raman scattering within the fiber at the wavelength specified.

Solution: The threshold optical power for SBS is given by

$$\begin{aligned} P_B &= 4.4 \times 10^{-3} d^2 \lambda^2 \alpha_{dB} \nu \\ &= 4.4 \times 10^{-3} \times 6^2 \times 1.3^2 \times 0.5 \times 0.6 \\ &= 80.3 \text{ mW} \end{aligned}$$

The threshold optical power for SRS is given by

$$\begin{aligned} P_R &= 5.9 \times 10^{-2} d^2 \lambda \alpha_{dB} \\ &= 5.9 \times 10^{-2} \times 6^2 \times 1.3 \times 0.5 \\ &= 1.38 \text{ W} \end{aligned}$$

Ref: "Optical Fiber Communications Principles and Practice"- John M. Senior, 3rd Edition.

Numerical-2

Observations:

The Brillouin threshold occurs at an optical power level of around 80 mW while the Raman threshold is approximately 17 times larger. It is therefore apparent that the losses introduced by nonlinear scattering may be avoided by use of a suitable optical signal level (i.e. working below the threshold optical powers).

However, it must be noted that the Brillouin threshold has been reported as occurring at optical powers as low as 10 mW in single-mode fibers. Nevertheless, this is still a high power level for optical communications and may be easily avoided. SBS and SRS are not usually observed in multimode fibers because their relatively large core diameters make the threshold optical power levels extremely high. Moreover, it should be noted that the threshold optical powers for both these scattering mechanisms may be increased by suitable adjustment of the other parameters

Ref: "Optical Fiber Communications Principles and Practice"- John M. Senior, 3rd Edition.

Fiber Fused Coupler

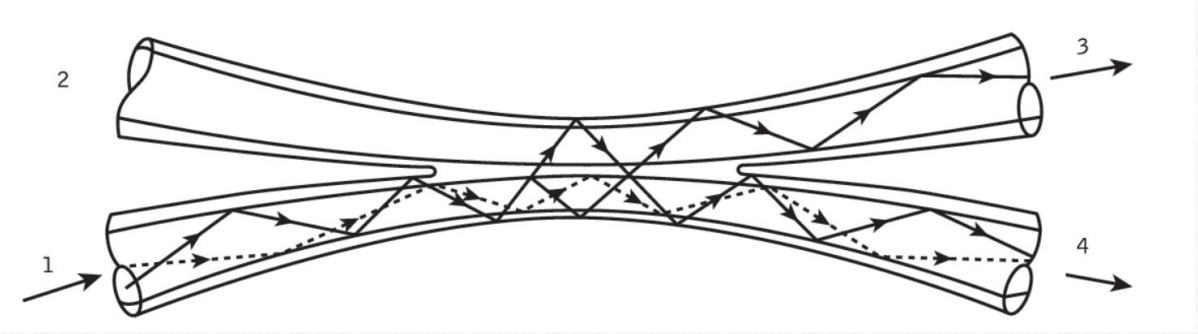


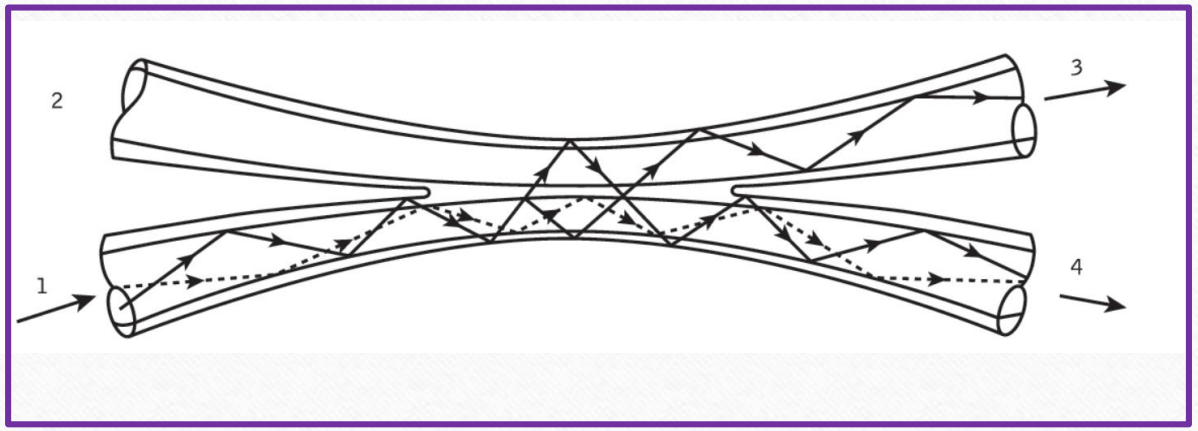
Figure : Structure and principle of operation for the fiber fused coupler (biconical taper)

The excess loss which is defined as the ratio of power input to power output is given by:

$$\text{Excess loss (four-port coupler)} = 10 \log_{10} \frac{P_1}{(P_3 + P_4)} \text{ (dB)}$$

Ref: "Optical Fiber Communications Principles and Practice"- John M. Senior, 3rd Edition.

The insertion loss, however, is generally defined as the loss obtained for a particular port to- port optical path. Therefore, considering Figure



$$\text{Insertion loss (ports 1 to 4)} = 10 \log_{10} \frac{P_1}{P_4} \text{ (dB)}$$

The crosstalk which provides a measure of the directional isolation† achieved by the device is the ratio of the backscattered power received at the second input port to the input power which may be written as:

$$\text{Crosstalk (four-port coupler)} = 10 \log_{10} \frac{P_2}{P_1} \text{ (dB)}$$

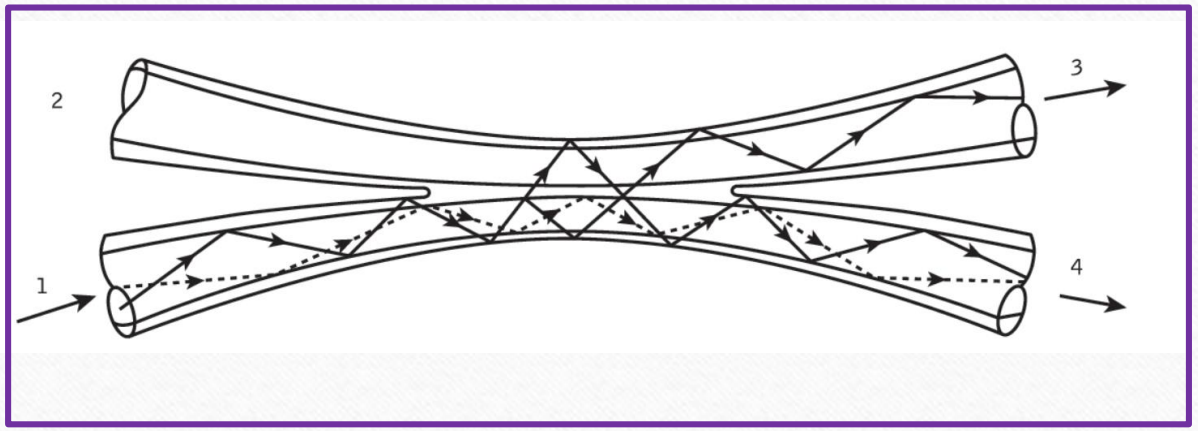
Note: The directional isolation and the crosstalk associated with a coupler are the same value in decibels but the former parameter is normally given as a positive value whereas the latter is a negative value.
Sometimes the directional isolation is referred to as the insertion loss between the two particular ports of the coupler which would be ports 1 to 2

Ref: "Optical Fiber Communications Principles and Practice"- John M. Senior, 3rd Edition.

Finally, the splitting or coupling ratio indicates the percentage division of optical power between the output ports. Again referring to Figure

$$\text{Split ratio} = \left[\frac{P_3}{(P_3 + P_4)}\right] \times 100\%$$

$$= \left[1 - \frac{P_4}{(P_3 + P_4)}\right] \times 100\%$$

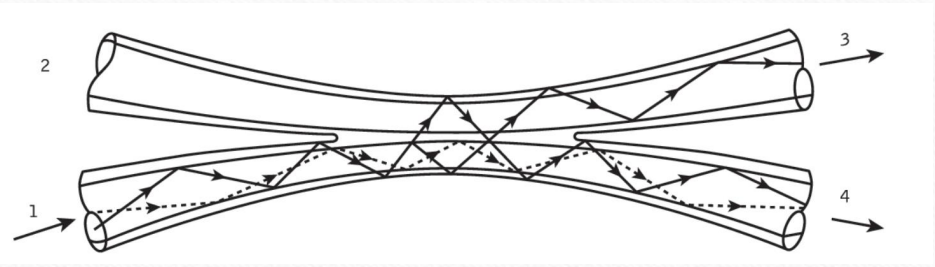


Ref: "Optical Fiber Communications Principles and Practice"- John M. Senior, 3rd Edition.

Numerical-3

A four-port multimode fiber FBT coupler has 60 μW optical power launched into port 1. The measured output powers at ports 2, 3 and 4 are 0.004, 26.0 and 27.5 μW respectively. Determine

(a) the excess loss,
(b) the insertion losses between the input and output ports,
(c) the crosstalk and
(d) the split ratio for the device.



Solution:

(a) The excess loss for the coupler may be obtained from

$$\text{Excess loss} = 10\log_{10}\frac{P_1}{(P_3+P_4)} = 10\log_{10}\frac{60}{53.5}$$

$$= 0.5\ \text{dB}$$

Ref: "Optical Fiber Communications Principles and Practice"- John M. Senior, 3rd Edition.

Numerical-3

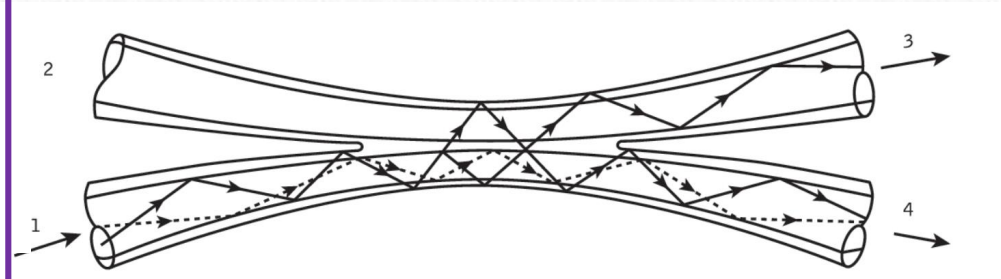
A four-port multimode fiber FBT coupler has 60 μW optical power launched into port 1. The measured output powers at ports 2, 3 and 4 are 0.004, 26.0 and 27.5 μW respectively. Determine the excess loss, the insertion losses between the input and output ports, the crosstalk and the split ratio for the device.

Solution:

(b) The insertion loss is provided by

$$\text{Insertion loss (ports 1 to 3)} = 10\log_{10}\frac{P_1}{P_3} = 10\log_{10}\frac{60}{26}$$

$$= 3.63 \text{ dB}$$

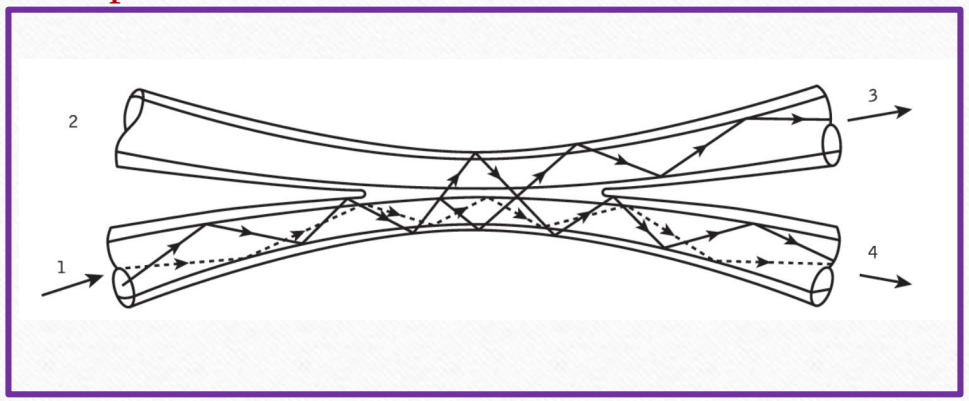


$$\text{Insertion loss (ports 1 to 4)} = 10\log_{10}\frac{60}{27.5} = 3.39 \text{ dB}$$

Ref: "Optical Fiber Communications Principles and Practice"- John M. Senior, 3rd Edition.

Numerical-3

A four-port multimode fiber FBT coupler has 60 μW optical power launched into port 1. The measured output powers at ports 2, 3 and 4 are 0.004, 26.0 and 27.5 μW respectively. Determine the excess loss, the insertion losses between the input and output ports, the crosstalk and the split ratio for the device.



Solution:

(c) Crosstalk is given by

$$\text{Crosstalk} = 10\log_{10}\frac{P_2}{P_1} = 10\log_{10}\frac{0.004}{60} = -41.8\ \text{dB}$$

(d) Finally, the split ratio can be obtained from

$$\text{Split ratio} = \left[\frac{P_3}{(P_3 + P_4)} \times 100\right] = \frac{26}{53.5} \times 100 = 48.6\%$$

Ref: "Optical Fiber Communications Principles and Practice"- John M. Senior, 3rd Edition.

To be contd……